Problem Statement

SiriusAI

ISOLATION FOREST

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WITH CODES

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AML & the Role of Anomaly Detection

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ExtraTreesClassifier
ExtraTreesRegressor
GradientBoostingClassifier
GradientBoostingRegressor
HistGradientBoostingClassifier
HistGradientBoostingRegressor
IsolationForest
RandomForestClassifier
RandomForestRegressor

API Reference > sklearn.ensemble > IsolationForest

IsolationForest

```
class sklearn.ensemble.IsolationForest(*, n_estimators=100, max_samples='auto', contamination='auto', max_features=1.0, bootstrap=False, n_jobs=None, random_state=None, verbose=0, warm_start=False) #
```

[source]

Parameters

n_estimators: int, default=100

Why It Matters:

- Each tree helps randomly isolate data points.
- The model calculates the average path length across all trees to determine if a point is anomalous.

Value	Behavior
Small (e.g. 10)	Fast training, but less accurate scores
Medium (100)	Good balance (default)
Large (500+)	More stable results, but slower

More trees → better generalization, but also more computation.

max_samples

Each tree is trained on a subset of your data.
Smaller subsets → faster and help detect rare outliers.

Value Type	What Happens
"auto"	Use min(256, total samples) → safe & fast default
int (e.g. 100)	Use exactly 100 samples for each tree
float (e.g. 0.5)	Use 50% of the dataset (0.5 * total samples) per tree

contamination

contamination = 'auto' or float (e.g., 0.05)

The model what fraction of the data is expected to be anomalies (outliers).
It is used to set the threshold for labeling points as:
-1 → anomaly
1 → normal

Value	What it means
'auto'	Let the model decide threshold (based on data, good for unknown cases)
float (e.g., 0.1)	You tell the model: “Expect 10% of the data to be outliers”

max_features

max_features = int or float, default=1.0

Type	What it does
int	Use exactly that number of features (e.g., max_features=3)
float	Use a fraction of total features (e.g., 0.5 = 50% of all features)
1	Use **all** features (default and fastest)

bootstrap & n_jobs

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bootstrap

bool, default=False
bootstrap: If True, trees use data with replacement (some rows can repeat); if False (default), each row is used only once per tree.

n_jobs:int, default=None

Number of CPU cores to use; None = 1 core, -1 = use all available cores.

random_state , verbose & warm_start

random_state : int,

Random_State instance or None, default=None
Controls randomness in selecting features and split points for each tree.
Set an int (e.g., 42) to get the same results every time you run the model.

verbose : int, default=0

Controls the verbosity of the tree building process. Controls how much progress info is printed while building trees — 0 means silent, higher numbers show more details.

warm_start: bool, default=False

When set to True, reuse the solution of the previous call to fit and add more estimators to the ensemble, otherwise, just fit a whole new forest.